

vating the scheduler. The kernel is then swapped out of the memory in lieu of the scheduler, except for some important library functions, and will start executing the child tasks.

Scheduling

The smallest unit of execution in an RTOS is called a task. A real time task is generated in response to an event. This event can be an internal event like a clock interrupt generated by a hardware clock every few milliseconds or an external interrupt generated when a predetermined event occurs like a sensor value going above or below a threshold. Nearly all real time systems consists of multiple real time tasks. The time constraints on different tasks will be different. It is onus on the scheduler and the operating system as a whole to make sure that the tasks are executed on time.

In this section we'll outline some fundamental task scheduling algorithms. It'll not be possible to discuss the nitty-gritties of each algorithm due the limitation of space but we'll list some resources at the end of this section, for those of you who want to dig in deeper.

The schedulers are broadly categorised into 2 main categories – clock driven and event driven. We'll discuss some of the important members of these broad classifying categories.

Clock driven

Clock driven schedulers are those for which the scheduling points occur only at timer interrupts. They're also known as off-line schedulers as the schedule is fixed before the system starts running, in other words the scheduler pre-determines which task will run next. As a result such schedulers incur very negligible runtime overhead with regards to making a decision for which task to schedule next. But due to their intransient nature such schedulers are ill-equipped to handle dynamically generated tasks. We'll briefly discuss the two important clock driven schedulers

Table driven

Table driven schedulers usually pre-determine which task to run next and store the result in a table at the time the system is designed or configured. The schedule is not automatically generated by the scheduler rather the application programmer is completely in control of the schedule and is free to decide the order in which the tasks will be executed and store it in the schedule table.